

UNIVERSITY OF BOTSWANA

2009/2010 – EXAMINATIONS

FRONT PAGE

COURSE NUMBER MAT111 DURATION 2 hrs DATE NOV 2009

TITLE OF PAPER INTRODUCTORY MATHEMATICS I

SUBJECT MATHEMATICS TITLE OF EXAMINATION BSc I

MORNING/AFTERNOON

INSTRUCTIONS:

- ANSWER ALL QUESTIONS IN SECTION A AND ANY TWO(2) QUESTIONS FROM SECTION B.
- ALL MARKS ARE INDICATED IN BRACKETS [].

NUMBER OF PAGES INCLUDING COVER PAGE: 5

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Section A

Answer All Questions: Each Question carries 10 Marks

Question 1

- (a) Write $0.\overline{513}$ in the form $\frac{p}{q}$, where $p, q \in \mathbb{Z}$ and $q \neq 0$. [3]
- (b) Solve the equation, $2^{2x}4^{x+1} = 32$ [3]
- (c) Solve the equation, $4^{x-1} = 3^{x+1}$ [4]

Question 2

Solve the following inequalities.

- (a) $|\frac{3}{4}x - 1| \leq \frac{1}{2}$. [4]
- (b) $x(x+1) > -2(2x+3)$. [6]

Question 3

- (a) Given that $f(x) = \frac{1}{x}$, $g(x) = 3x$ find $(g \circ f)(x)$ and state its domain and range. [6]
- (b) Find the equation of the line passing through $(-1, -2)$ and perpendicular to $2x - y + 1 = 0$ [4]

Question 4

- (a) Find the value of k if the remainder, when the polynomial

$$P(x) = 2x^4 + kx^3 - 11x^2 + 4x + 12$$

is divided by $x - 3$, is 60. [5]

- (b) Determine the polar form of the following complex number $-1 - \sqrt{3}i$ [5]

Question 5

(a) Evaluate $\cos \frac{3\pi}{4}$ using a double angle identity. [6]

(b) Evaluate $\tan^{-1}(\sqrt{3})$ without using a calculator. [4]

Question 6

Compute

$$(1 + i)^5 \quad [10]$$

Section B

Answer TWO(2) Questions OUT of FOUR:
Each Question carries 20 Marks

Question 7

- (a) Find rational numbers a and b such that

$$\sqrt{7 + 2\sqrt{10}} = \sqrt{a} + \sqrt{b}$$

[6]

- (b) Solve the equation,

$$4^{x+1} + 15(4^{-x}) - 16 = 0$$

[8]

- (c) If α and β are the roots of equation $x^2 + 3x + 1$, find the equation whose roots are 2α and 2β .

[6]

Question 8

- (a) Solve the inequality: $4 \leq \left| \frac{1}{3}x + 4 \right| \leq 6$.

[10]

- (b) Given that

$$f(x) = \sqrt{x}, \quad g(x) = 1 - x^2$$

Find $f \circ g$ and find its domain and range.

[10]

Question 9

- (a) Given the function

$$f(x) = x^2 - 4x + 3,$$

find (i) the y-intercepts, (ii) the x-intercepts, (iii) the vertex, and (iv) Sketch the graph of f . Hence or otherwise sketch the graph of

$$g(x) = (x - 1)^2 - 4(x - 1) + 3.$$

[12]

- (b) When a polynomial $P(x)$ is divided by x the remainder is -1 . When $P(x)$ is divided by $x + 1$ the remainder is 4 . When $P(x)$ is divided by $x - 2$ the remainder is 1 . Find the remainder when $P(x)$ is divided by $x^3 - x^2 - 2x$.

[8]

Question 10

- (a) Evaluate $\cos 4x$ given that

$$\sin x = \frac{\sqrt{3}}{2}$$

[5]

- (b) Prove

$$\cos 2x = \frac{1 - \tan^2 x}{\sec^2 x}$$

[5]

- (c) Find all solutions of the following trigonometric equation

$$\sin 2x = \sin x$$

[10]